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Improved CNN architecture for automated classification of skin diseases

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ABSTRACT

Rosacea, atopic dermatitis and bullous disease are significant skin conditions with distinct characteristics and impacts, which affect millions of people globally. Early and quick diagnosis is crucial for these conditions to prevent complications, alleviate symptoms and improve quality of life for affected individuals. This research article presents an innovative approach to automated classification of common skin diseases using Convolutional Neural Network (CNN) models. The study focuses on diagnosing rosacea, atopic dermatitis and bullous disease, leveraging CNN technology. Four pre-trained CNN models – DarkNet-53, ResNet-18, SqueezeNet and EfficientNet-b0 – were investigated. Additionally, two improvised versions of DarkNet-53 and an improvised version of ResNet-18 were developed, integrating a specific number of fully connected neural network layers and adjusting other CNN layers such as batch normalisation and activation function layers accordingly. The performance of these improvised models surpassed that of the original architectures, demonstrating superior accuracy in identifying the targeted skin diseases. In terms of overall accuracy, improvised versions of DarkNet-53 and ResNet-18 achieved an accuracy of 98% and 98%, respectively, while the original models could achieve an accuracy between 75% and 80%. This research contributes to advancing automated diagnostic systems for dermatological conditions, potentially improving early detection and treatment outcomes.

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KEYWORDS
Skin; rosacea; atopic dermatitis; bullous disease; CNN

1. Introduction

Rosacea (Acne Rosacea), atopic dermatitis and bullous disease are three of the most common skin conditions that affect millions of people worldwide. Rosacea is chronic, causing facial redness and can result in eye problems if left untreated for a considerable amount of time (Wilkin 1994; Huynh et al. 2022). Atopic dermatitis, including eczema, causes skin inflammation and intense itching, impacting daily life and sleep quality (Berke et al. 2012). Bullous diseases, characterised by blister and bullae formations, range from mild to life-threatening conditions, such as *pemphigus vulgaris* (Yang et al. 2019). These skin diseases not only affect physical health but also have profound psychological and social consequences. Rosacea, in particular, can lead to low self-esteem and depression (Huynh et al. 2022). Atopic dermatitis can cause embarrassment and social withdrawal due to visible symptoms (Wilkin 1994). Severe bullous diseases can be life-threatening, requiring prompt diagnosis and treatment (Berke et al. 2012).

In this context, development of automated skin disease identification systems using biomedical image analysis can play a critical role in their containment and treatment. Early detection and intervention can prevent disease progression and complications by facilitating timely treatment decisions, improving patient outcomes and quality of life (Ahammed et al. 2022). Scalability and efficiency enable healthcare professionals to manage a large volume of cases, reducing diagnostic delays.

Machine learning techniques have emerged as powerful tools for the automatic identification of various skin diseases. Studies have shown the effectiveness of deep convolutional neural networks (CNNs) in detecting acne lesions with high accuracy rates (Junayed et al. 2019, 2021). Deep CNNs have demonstrated the capability to identify rosacea lesions and distinguish them from other skin diseases (Zhao et al. 2021). Machine learning methods combined with skin physiology analysis has been found to contribute to the differential diagnosis of rosacea (Ge et al. 2022). Computer-aided diagnosis using deep learning models has shown promise in accurately classifying various types of dermatitis based on image features (Bajwa et al. 2020). It is observed from the literature that there is a lack of specific studies focusing on machine learning techniques for bullous disease identification. However, the general applicability of deep learning approaches in skin disease classification suggests its potential utility in this domain.

Houssein et al. (2022) provides a detailed overview of the deep and machine learning techniques for image-based skin disease identification. Broadly, two approaches have been used for image-based skin disease identification. They are image segmentation (Houssein et al. 2021, 2022 Nassef et al. 2022, 2023; Emam et al. 2023) and classification (Houssein et al. 2022, 2022; Houssein and Sayed 2023). Table 1 lists and discusses the adopted techniques, datasets used to train the models, the diseases being identified, and the achieved results in some recent studies in literature on the automated classification of skin diseases.

From the literature review, it is found that there exist limited studies on classification of rosacea, atopic dermatitis and

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Table 1. Comparison of recent studies on CNN model development for skin disease identification.

Reference	Model	Dataset	Results
Binol et al. (2020)	Inception-ResNet-v2 and ResNet-101	41 clinical images collected at the Ohio State University, USA for rosacea detection.	Average dice coefficient: 0.898 (Inception-ResNet-v2) and 0.878 (ResNet-101)
Zhao et al. (2021)	ResNet-50	24,736 photos among which 18,647 were related to rosacea and remaining to acne, facial seborrheic dermatitis and eczema. Target disease being rosacea.	Overall accuracy and precision of 75.7% and 66.7%, respectively, for differentiation between rosacea, seborrheic dermatitis and eczema.
Park et al. (2023)	19 different pre-trained CNNs	Clinical images from 40 patients with rosacea and 59 from control group collected at Johns Hopkins University Hospital, USA	Overall accuracy of 92.27% using the top five CNNs by employing the majority rule.
Junayed et al. (2020)	InceptionV3, MobileNetV1 and EczemaNet	500 images related to five different types of dermatitis collected from different various hospitals and public sources.	Overall accuracy of 96.2%
Padilla et al. (2019)	MobileNet	6,264 images collected from (Liao et al. 2016) related to atopic dermatitis and psoriasis chronic plaque.	Overall accuracy of 88% and 90%, respectively, for atopic dermatitis and psoriasis chronic plaque.
Pan et al. (2020)	InceptionV3, MobileNet, ResNet-50, VGG-16, VGG-19, EczemaNet	3926 images based on original dataset from (Thomas et al. 2011) for grading the severity of dermatitis.	EczemaNet performed with least RMSE of 1.929 ±0.019.
Bang et al. (2021)	12 different pre-trained CNN models	24,852 clinical images collected from Seoul St. Mary's Hospital consisting of different grades of atopic dermatitis.	Overall accuracy ≥96% for all diseases
Shi et al. (2021)	9 different pre-trained CNN models	220,000 direct immunofluorescence microscopy images of bullous disease collected by the authors.	A modified version of GoogLeNet gave the best accuracy of 89.3%.
Swain et al. (2020)	ConvNet model	1,600 images from Dermnet database pertaining to six diseases – acne rosacea, bullous, cellulitis impetigo and bacterial infections, eczema, melanoma and nail fungus.	Overall accuracy of 93.06% for a test dataset of 180 images

bullous disease CNN models. Moreover, improvisations of pre-trained CNNs are an area with immense potential to achieve better predictive performance. In this regard, the present study aims to compare four well-known pre-trained CNNs viz., SqueezeNet, DarkNet-53, EfficientNet-b0 and ResNet-18 with two modified versions of DarkNet-53 and a modified version of ResNet-18. This investigation will attempt to evaluate the effect of modifying the architecture of deep learning models in altering their performance. The capabilities of the seven CNN models will be compared based on standard metrics of multi-classification obtained from the confusion matrix including accuracy, precision, recall, specificity and F1-score (Deepak and Bhat 2023a, 2023b, 2024). Also, the receiver operating characteristics (ROC) curve is plotted to assess the reliability of the model results.

2. Methodology

The present work consists of image preprocessing, followed by the definition of deep learning model training parameters. Then the trained model is used to classify the skin image into the three disease categories. This overall workflow of the study is depicted in Figure 1

2.1. Dataset and pre-processing

This research work uses the Dermnet data set available opensource at Kaggle repository: https://www.kaggle.com/datasets/shubhamgoel27/dermnet. The dataset contains photographs of more than 20 types of skin diseases in JPEG format consisting of RGB layers. The data set contains 311 images of rosacea, 123 images of atopic dermatitis and 113 pertaining to bullous disease. To maintain the

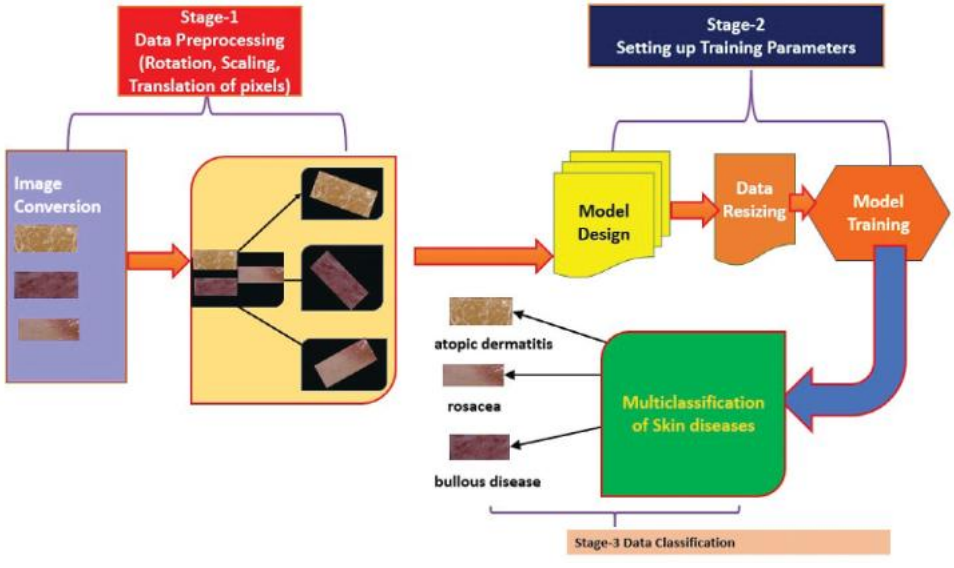


Figure 1. Workflow of present study.

balance between datasets, in this study, 100 images were taken randomly from each category of skin diseases from the datasets. A challenge which was faced in using this dataset is the presence of aspects unrelated to skin disease such as facial structures and body parts of the patient. Therefore, initial pre-processing was conducted in terms of cropping the images to obtain only the relevant information related to skin diseases from them.

2.2. Data augmentation

To increase the size of the dataset to train the models, three types of image manipulations were done for data augmentation – images were rotated (by 30 and 75 degrees), scaled (by factors of 1.4 and 1.8) and by translating the images (shifting by 10 to 70 pixels). Sample images after data augmentation processes are shown in Figures 2, 3 and 4. These augmentation techniques resulted in an increase in the dataset size by 10%.

2.3. Convolutional neural networks

Four pre-trained CNNs and three improvised versions were investigated in this study. Their configurations and important features are described in this section.

2.3.1. SqueezeNet

The fire-protection module (Figure 5) is an important part of the SqueezeNet CNN. It reduces the network's hyperparameters by switching the 3×3 filter for a 1×1 filter, which thereby reduces

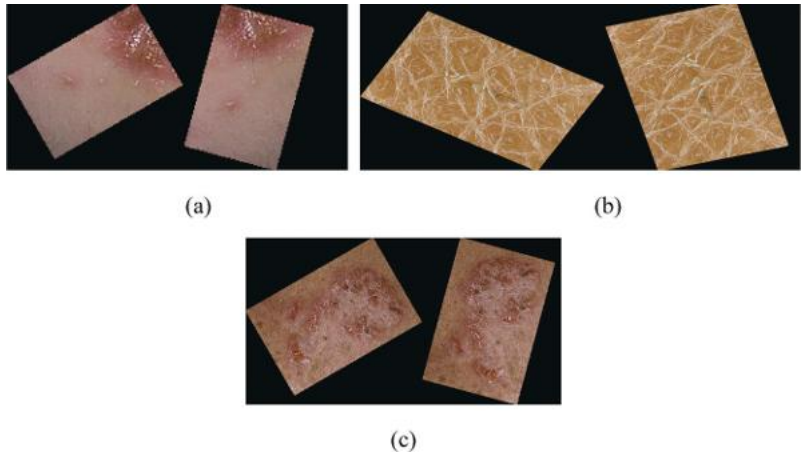


Figure 2. Data augmentation by rotation: (a) rosacea, (b) atopic dermatitis, (c) bullous disease.

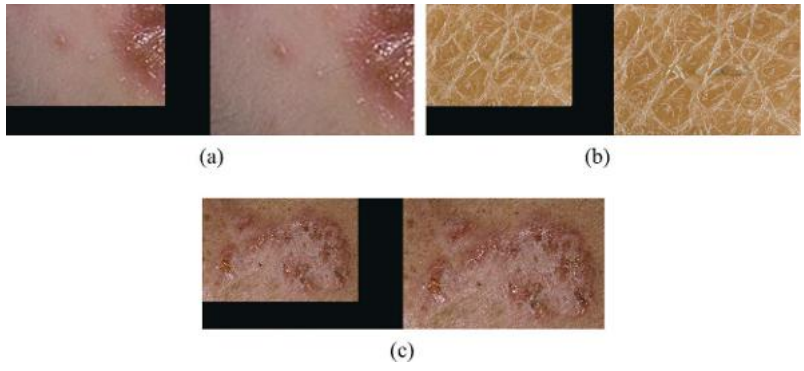


Figure 3. Data augmentation by scaling: (a) rosacea, (b) atopic dermatitis, (c) bullous disease.

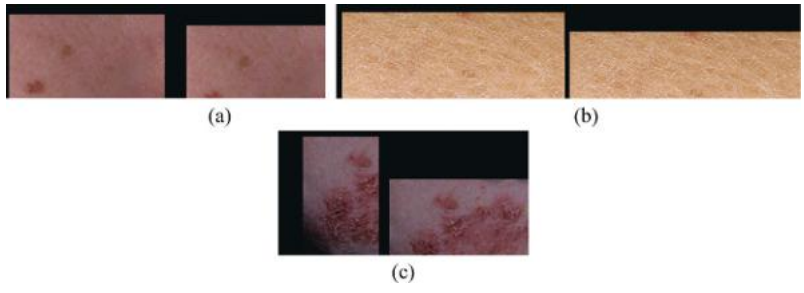


Figure 4. Data augmentation by translation: (a) rosacea, (b) atopic dermatitis, (c) bullous disease.

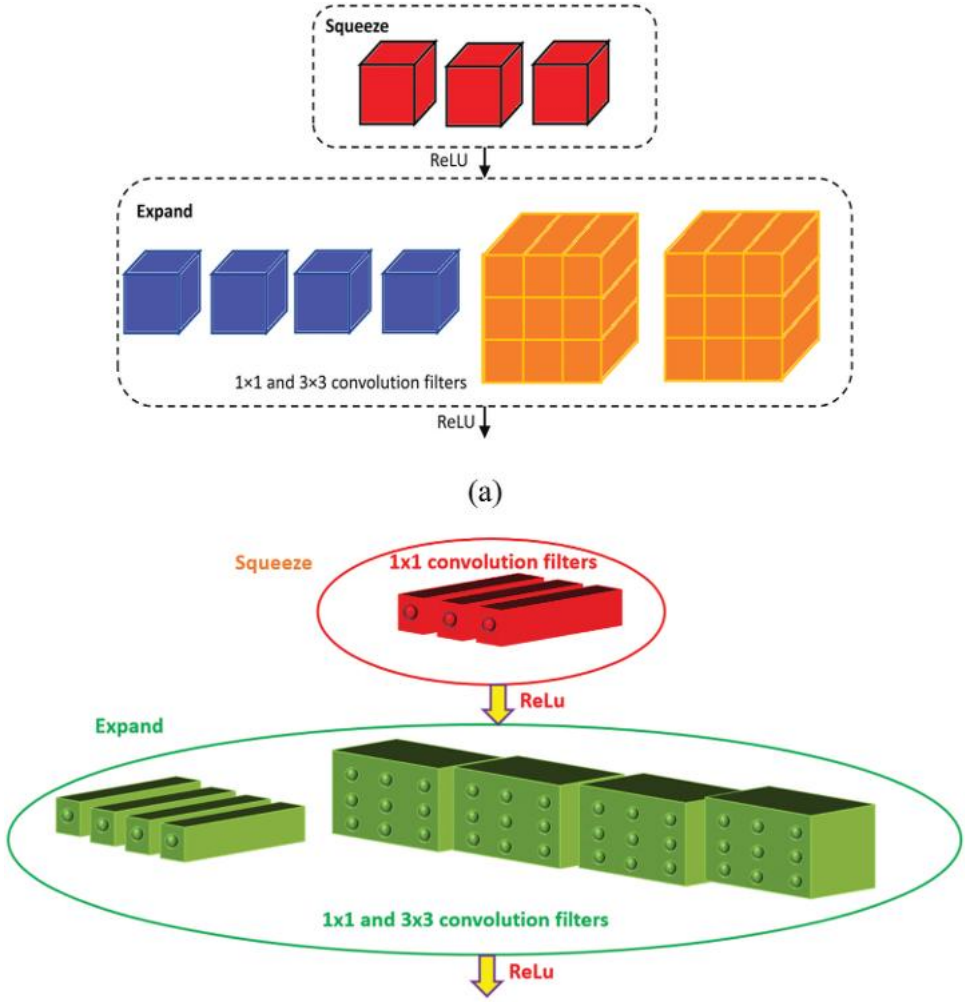


Figure 5. (a) Fire protection module in SqueezeNet, (b) microarchitectural view of fire module with organization of convolution filters. Here, s1×1 = 3, e1×1 = 4 and e3×3 = 4.

the input channels to the 3×3 filter. In order for the convolution layers to cover bigger activation maps, it maintains the down-sampling later in the network. Deep compression is utilised to scale the model. SqueezeNet compresses the modal volume by 510 times its original volume and employs 50 times less parameters than AlexNet, a widely used CNN model, without sacrificing accuracy (Iandola et al. 2016). The fire protection module comprises a squeeze convolution layer (having only 1×1 filters), that feeds into an expand layer which contains both 1×1 and 3×3 convolution filters; as shown in Figure 5a.

There are three tuneable dimensions (hyperparameters) in a fire module: s1 $\times$ 1, e1 $\times$ 1 and e3 $\times$ 3. In a Fire module, s1 $\times$ 1 is the number of filters in the squeeze layer (all 1×1), e1 $\times$ 1 is the number of 1×1 filters in the expand layer and e3 $\times$ 3 is the number of 3×3 filters in the expand layer (Iandola et al. 2016). Here, the convolution filters are illustrated in Figure 5b.

Figure 6 depicts the SqueezeNet network's overall architecture.

2.3.2. DarkNet-53

Because Darknet-53 is a comparatively deeper network, it faces the problem of gradient divergence, which is resolved by employing residual networks to facilitate effective network learning (He et al. 2016). Figure 7 presents the Darknet-53 schematic diagram. The number of residual components that are repeated is shown by the numbers 1, 2, 8 and 4 in the leftmost column. There are two convolutional layers in the residual connection. The first layer of convolution has a kernel size of 1×1, whereas the second layer has a kernel size of 3×3 (Figure 7a). The Cross Stage Partial Block (CSPBlock) employs a strategy to partition the feature map of the base layer into two parts and then merges them through a cross-stage hierarchy (Figure 7b).

Mish is a novel self-regulated non-monotonic activation function inspired by the self-gating property of Swish. Mish is mathematically defined as, $f(x) = x\tanh(\ln(1 + e^x))$. Mish is bounded below and unbounded above with a range of $[\approx -0.31, \infty)$. Mish uses the self-gating property where the non-modulated input is multiplied with the output of a non-linear function of the input. Due to the preservation of a small amount of negative information, Mish is eliminated by designing the preconditions necessary for the Dying ReLU phenomenon. This property enables better expressivity and information flow. Being unbounded above, Mish avoids saturation, which generally causes training to slowdown drastically due to near-zero gradients. Being bounded below is also advantageous

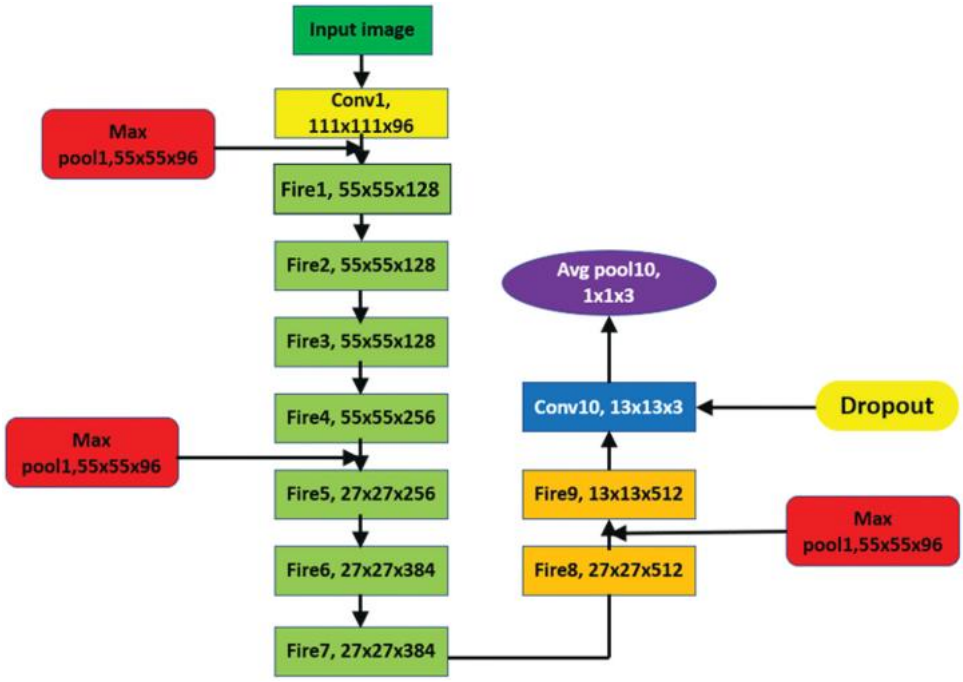


Figure 6. Overall architecture of SqueezeNet.

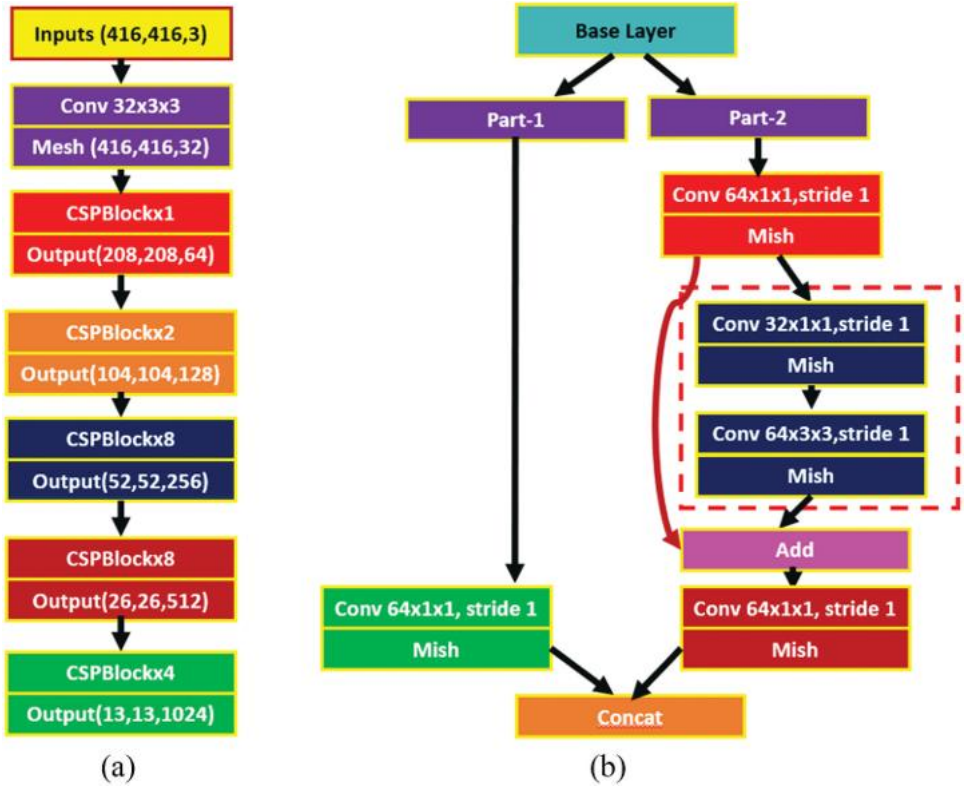


Figure 7. Schematic diagram of DarkNet-53: (a) overall and (b) CSPBlock architecture.

since it results in strong regularisation effects. Unlike ReLU, Mish is continuously differentiable, a property that is preferable because it avoids singularities and, therefore, undesired side effects when performing gradient-based optimisation (Misra 2019).

The intricate architecture of DarkNet-53 is presented in Figure 8. As the final model output, the network uses Rectified Linear Unit (ReLU) activation after adding output x to output $f(x)$ (Ioffe and Szegedy 2015). When the input x is passed directly to the output, the end outcome is $f(x) = x$;

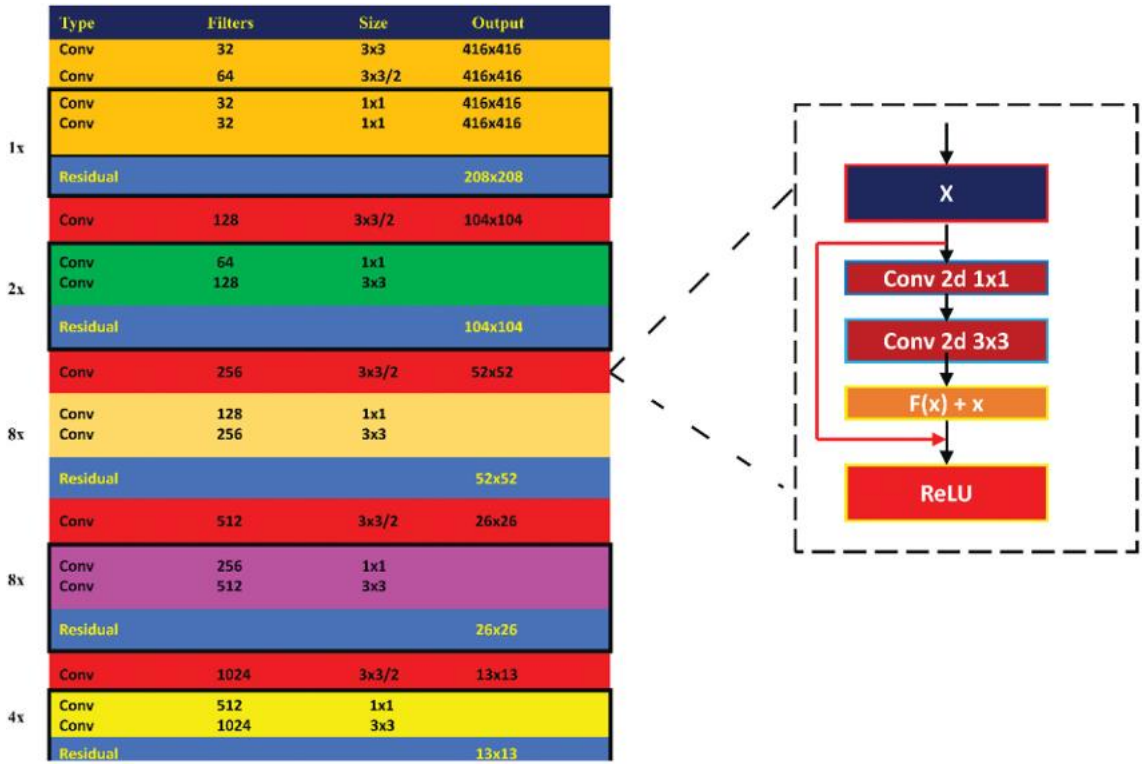


Figure 8. Architecture of DarkNet53 network.

when $f(x) = 0$ is reached, $H(x) = x$, the residual approaches 0 and the training model converges. With this structure, the network will converge even with many layers present, and training the model will be simple.

Hence, as the network becomes deeper, the expression of features is enhanced while it also maintains the recognition precision. Furthermore, the convolution 1×1 present in the residual network decreases the channel of each convolution, which consequently reduces the number of parameters. Also, the amount of computation is reduced to a significant extent which leads to the convergence of the model.

2.3.3. EfficientNet-b0

With 83 layers, EfficientNet-b0 is the deepest neural network considered in the present study. EfficientNet-b0 accepts input images with a resolution of 224 by 224 pixels (Tan and Le 2019). It is noted for implementing MobileNet-V2 equivalent Mobile Inverted Bottleneck Convolution (MBConv), albeit with a bigger budget for floating-point operations per second (FLOPs). When compared to the other CNN network models, EfficientNet-b0 is significantly more accurate and efficient. It also reduces FLOP and parameter sizes by an order of magnitude. EfficientNet-b0 employs Swish which is a product of sigmoid and linear activation functions whereas the other existing CNN models apply ReLU which acts as activation functions (Prajit et al. 2017; Hridoy et al. 2021; Deepak and Bhat 2023a, 2024). EfficientNet-b0 uses lesser trainable parameters as it deploys an inverted residual block. The overall architecture of EfficientNet-b0 is shown in Figure 9.

2.3.4. ResNet-18

ResNet-18 is a CNN architecture that belongs to the ResNet (Residual Network) family. Its architecture comprises 18 layers, including convolutional layers, pooling layers, fully connected layers and skip connections. These skip connections help mitigate the vanishing gradient problem, allowing for deeper networks while maintaining computational efficiency (refer Figure 10a) (He et al. 2016). The core of ResNet-18 is formed by residual blocks. Each residual block contains two convolutional layers with batch normalisation and ReLU activation functions. The pooling layers are utilised to reduce the spatial dimensions of the feature maps, aiding in translation invariance and reducing computational complexity. ResNet-18 ends with fully connected layers followed by a SoftMax activation function for classification tasks, mapping the extracted features to class probabilities (Figure 10b) (Wu et al. 2021; Gao et al. 2021; Oğuz and Ertuğrul 2023).

2.3.5. DarkNet-53-2FC3

Recently, studies have shown that modifications in the architecture of pre-trained models can have a significant influence on their performance (Pathak and Raju 2021, 2023; Jayapalan and Anuar 2022). The present study proposes two modified versions of DarkNet-53 CNN by adding fully connected layers which are used to extract the important features to identify the disease pathologies from the images. In the first variant named DarkNet-53-2FC3, two fully connected layers (FCL) were added to the original DarkNet-53 model by incorporating two sets of batch normalisation, swish and FCL layers.

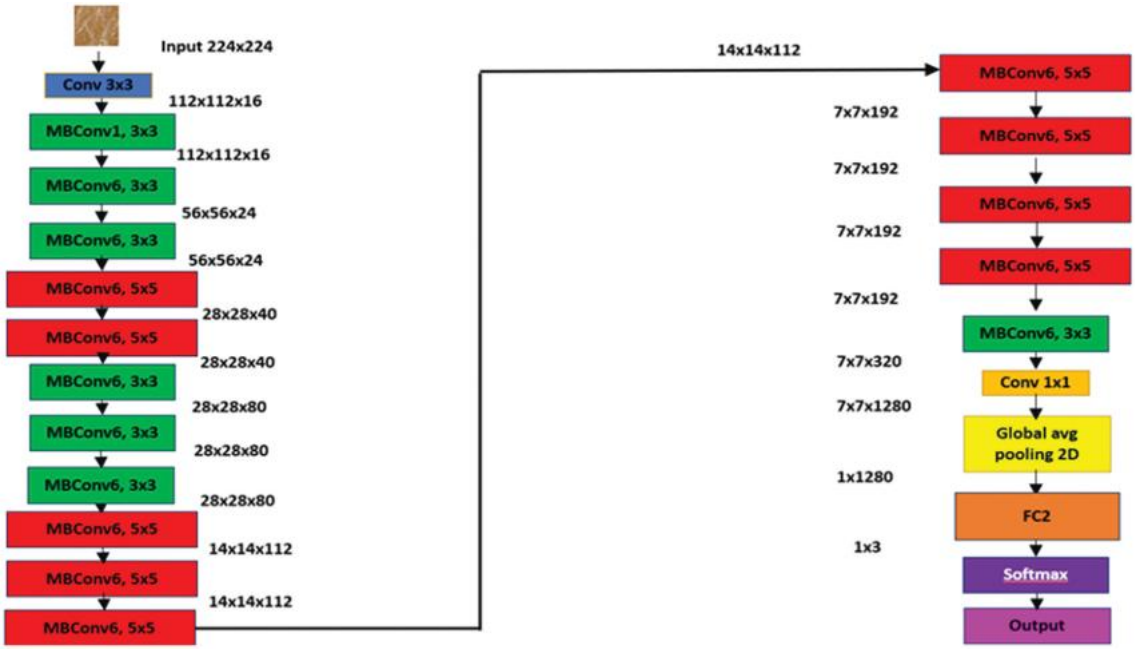


Figure 9. Overall architecture of EfficientNet-b0.

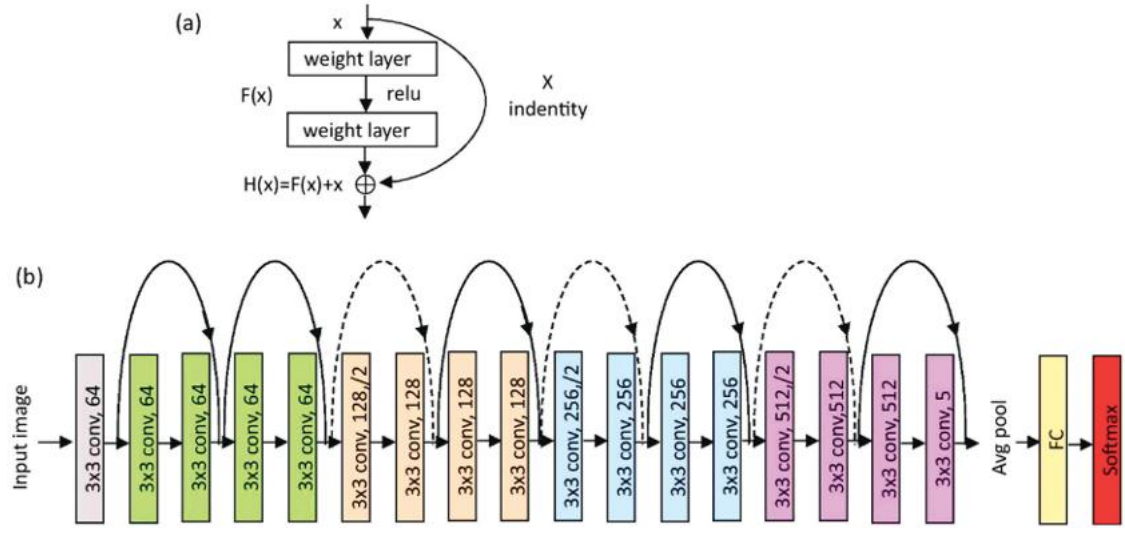


Figure 10. (a) Residual block and (b) overall architecture of ResNet-18.

Batch normalisation is a method of normalising activations in deep neural network intermediate layers. It helps the training process in saturated non-linearity regimes. The size of the first and second FCL are 512 and 3, respectively, which then seamlessly connects with the classification layer for three classes of diseases, viz., rosacea, atopic dermatitis and bullous diseases, which are being identified in the present work. The modified part of the DarkNet-53 architecture is shown in Figure 11.

2.3.6. DarkNet-53-4FC3

The second variant, named DarkNet-53-4FC3, incorporates four FCL with sizes: 2048, 1024, 512 and three number of neurons. To achieve this architecture combination, four sets of batch normalisation, swish and FCL layers are added to include the four FCL to the original DarkNet-53 model. The last FCL seamlessly connects with the classification layer with three outputs corresponding to the diseases being identified herein. The modified part of DarkNet-53 architecture is shown in Figure 12.

2.3.7. ResNet-18-2FC3

The ResNet-18 CNN architecture is modified by incorporating two FCL. This is done by including two sets of batch normalisation, ReLU and FCL layers, which finally connect to the SoftMax and finally the classification layer. This architecture is named the ResNet-18-2FC3 model signifying the two added FCL among

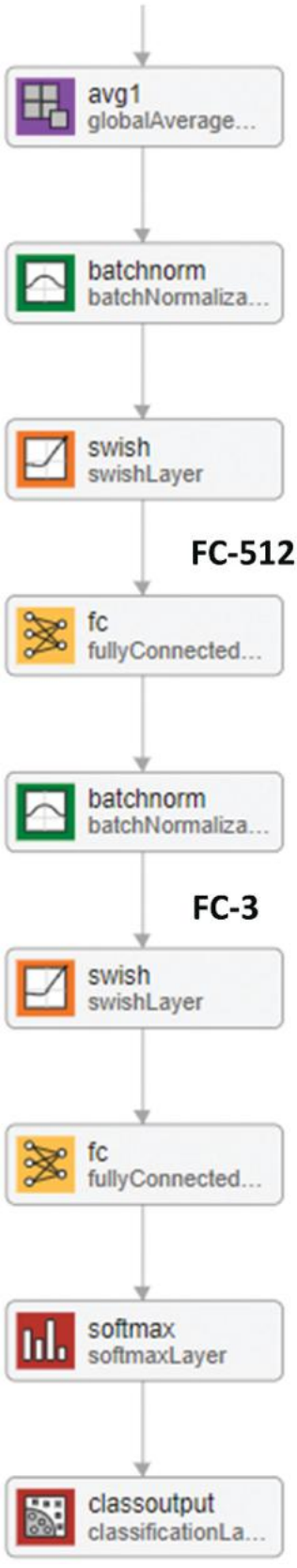


Figure 11. Modified DarkNet-53 with two additional FCL: FC-512 and FC-3.

which the final FCL is of size 3 to connect with the classification layer of the same size (refer Figure 13). The three modified CNNs are compared with the four original CNN models (SqueezeNet, DarkNet-53, EfficientNet-b0 and ResNet-18) discussed in this section.

3. Results and discussion

3.1. Accuracy of the CNN models

The trained models were tested for a random batch of 100 images consisting of 33, 33 and 34 images pertaining to rosacea, atopic dermatitis and bullous disease, respectively. The comparison of the overall classification accuracy and the important features such as input/output image size, layers and parameters of the four original CNNs are shown in Table 2. It is found that all four models show an increase in accuracy of 3–6% with data augmentation. The maximum accuracy obtained is 80% by all models except EfficientNet-b0 which showed the least accuracy of 75%. Table 3 shows the performance of the improvised CNNs. The positive effect of data augmentation is evident in this case as well. Importantly, DarkNet-53-4FC3 showed the best accuracy of 98% out of the three CNNs, followed by ResNet-18-2FC3 and DarkNet-53-2FC3 with 97% and 85%, respectively. Figures 14 and Figure 15 pictorially depict the classification accuracy of the original unmodified and modified CNNs, respectively, both with and without performing data augmentation operations. Figure 16 shows representative examples of the identification of skin diseases by the seven trained CNNs along with their confidence levels. The superior performance of modified versions of CNNs is also evident in the case of the confidence scores shown along with their disease predictions in Figure 16.

3.2. Performance metrics

The confusion matrix is crucial in evaluating the performance of CNNs in multiclassification problems. It provides a comprehensive overview of the model's predictive capabilities by summarising the classification results across different classes. With a confusion matrix, one can easily identify and compare where each CNN excels and where it struggles in classifying specific classes. Additionally, metrics derived from the confusion matrix, such as accuracy, precision, recall and F1-score, offer deeper insights into their performance behaviour (Deepak and Bhat 2023a, 2023b, 2024). Figure 17 shows the confusion matrices generated for each of the seven CNNs developed in this study for a set of 100 test images. It is observed that DarkNet-53-4FC3 and ResNet-18-2FC3 showed superior performance compared to the other models with an overall accuracy of 98% and 97%. This is clearly seen by the limited number of off-diagonal terms (refer Figure 17c,e).

Tables 4 and 5 show the performance metrics viz., accuracy, precision, recall, specificity and F1-scores for the original and

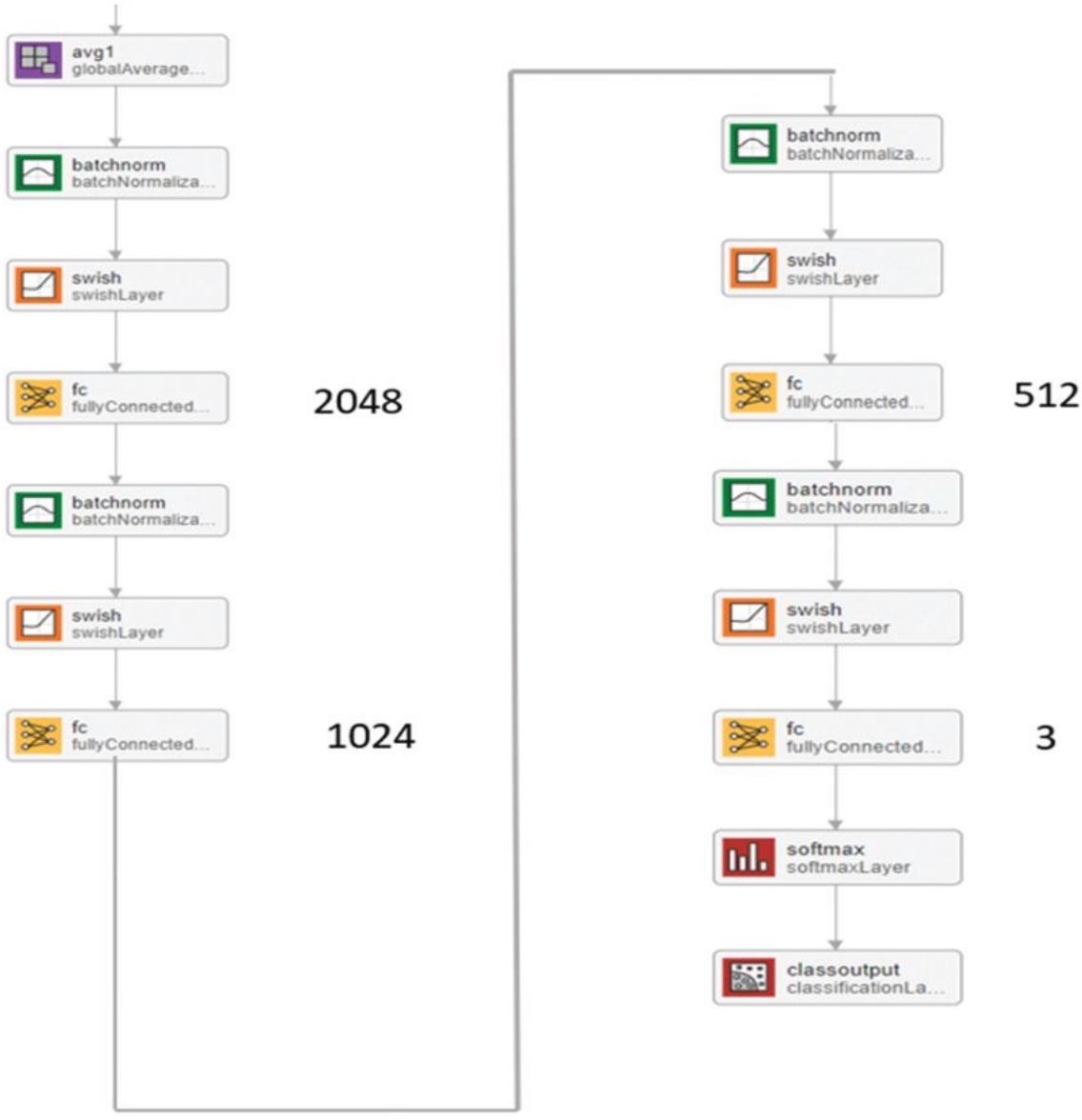


Figure 12. Modified DarkNet-53 with four additional FCL: FC-2048, FC-1024, FC-512 and FC-3.

modified CNNs, respectively. In addition to these metrics, the Matthews correlation coefficient (MCC) is computed which is beneficial to assess model performance in multi-class problems. More details about MCC can be found here (Anbalagan et al. 2023). These are critical parameters to assess the capability of a CNN in case of a multiclassification identification setting. Again, the key takeaway is the consistent good performance of DarkNet-53-4FC3 and ResNet-18-2FC3 across all the decision metrics, highlighting their suitability for the skin disease identification problem. The addition of FC layers has a strong influence on the model performance as shown recently by Basha et al. (2020). The present results also show that the modification of FC layers can have a significant influence on model performance. The maximum accuracy shown is comparable to the performance achieved by (Padilla et al. 2019; Junayed et al. 2020; Swain et al. 2020; Bang et al. 2021; Shi et al. 2021; Park et al. 2023). Figure 18 summarises the performance metrics shown in Table 4 by means of a bar graph for all the seven different CNNs evaluated herein.

ROC (Receiver Operating Characteristic) curves are vital in assessing the performance of CNNs in multiclassification problems. They provide a comprehensive visualisation of the model's ability to discriminate between different classes by plotting the trade-off between true positive rate (sensitivity) and false positive rate (1/specificity) across various decision thresholds. This allows for a nuanced understanding of CNN's classification capabilities, especially in scenarios where multiple classes are involved. ROC curves facilitate comparison between different

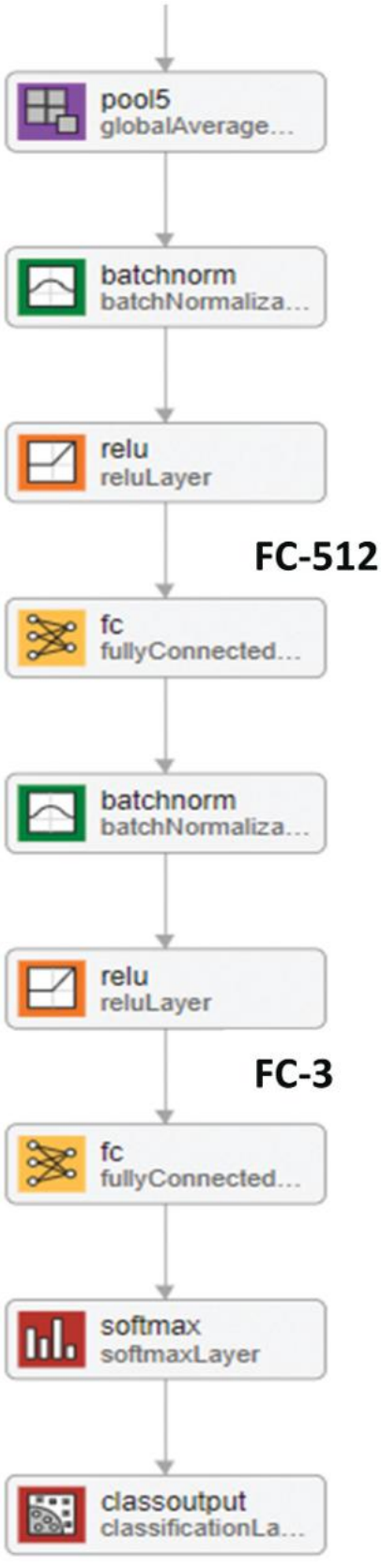


Figure 13. Modified ResNet-18 with two additional FCL: FC-512 and FC-3.

models and help identify the optimal balance between true positives and false positives, aiding in model selection and performance optimisation (Choi et al. 2014; Janssens and Martens 2020; Inthiyaz et al. 2023). Figure 19 shows the ROC for the three improvised CNNs. Again, it is clearly observed that DarkNet-53-4FC3 and ResNet-18-2FC3 have shown excellent capability for the automated identification of skin diseases considered in this study (rosacea, atopic dermatitis and bullous disease).

Although the dataset used in the present work included 20 types of skin diseases, only three of the most prevalent skin disease categories were considered here. The future scope of the present work is possible extension to other types of diseases. Several studies in literature have shown that an ensemble of networks can improve the robustness of ML and deep learning models (Keerthana et al. 2023; Elangovan et al. 2024; Vijayalakshmi et al. 2024). While the current study employs only four different CNNs in the analysis, utilising the ensemble techniques with a larger variety of models will be considered in the future.

Image preprocessing through image contrast enhancement techniques can greatly improve classification performance, as shown in recent studies by Vijayalakshmi and Nath (2022, 2023), which has not been considered in this work. In subsequent studies, this aspect needs to be considered to improve the model's performance. The present work did not propose a new backbone CNN model, but rather modified the available pre-trained models through the addition of FC layers. In future work, it is aimed to developed fundamental backbone models for such challenging tasks.

4. Conclusions

The present study developed seven CNN models for the automated identification of three of the common skin diseases viz., rosacea, atopic dermatitis and bullous disease. Four pre-trained models – DarkNet-53, ResNet-18, SqueezeNet, EfficientNet-b0, and three modified versions of the CNNs – DarkNet-53-2FC3, DarkNet-53-4FC3 and ResNet-18-2FC3 were trained to identify the three categories of skin diseases. The modified CNNs incorporated multiple levels of fully connected layers to enhance the feature-learning capabilities. From the performance analysis, it is found that DarkNet-53-4FC3 and ResNet-18-2FC3 showed the best performance with overall accuracy of 98% and 97%, respectively. The results from the present study highlight the significant influence of modifications in pre-trained models in improving their potential for identifying complex patterns such as skin diseases. The future scope of the work is extending the models to identify more categories of skin diseases. To this end, ensemble of networks can be utilised. Further, techniques such as contrast enhancement can further improve the reliability of the model predictions. Also, testing and

Table 2. Accuracy and important features of four original CNNs.

Network	Accuracy % (With data augmentation)	Accuracy % (Without data augmentation)	Input Image Size	Output Image Size	Number of layers	Parameters (millions)
SqueezeNet	80	74	227×227	801×826	18	1.24
Darknet-53	80	77	256×256	436×359	53	8.8
EfficientNet-b0	75	70	224×224	1920×926	82	5.3
ResNet-18	80	76	224×224	797×823	18	11.7

Table 3. Accuracy and important features of three modified CNNs.

Network	Accuracy % (With data augmentation)	Accuracy % (Without data augmentation)	Input Image Size	Output Image Size	Number of layers	Parameters (millions)
DarkNet-53-2FC3	85	80	256×256	436×359	55	8.8
DarkNet-53-4FC3	98	95	256×256	436×359	57	8.8
ResNet-18-2FC3	97	93	224×224	797×823	20	11.7

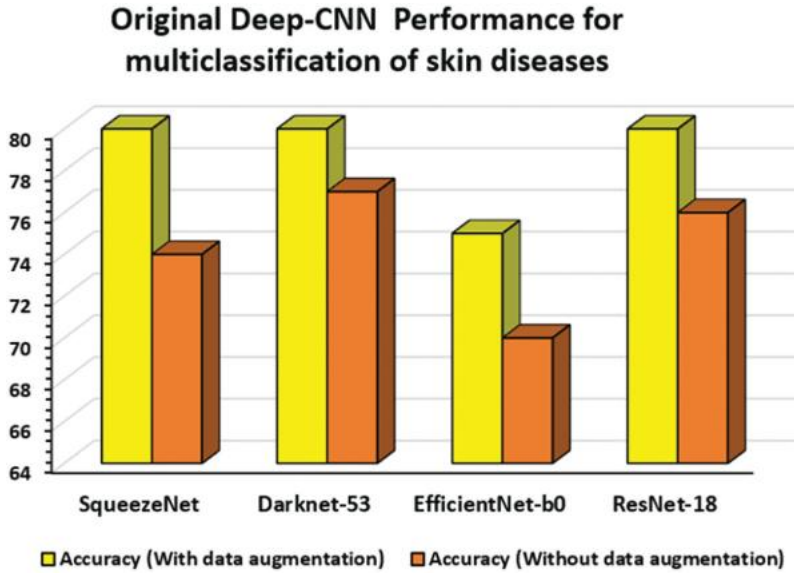


Figure 14. Performance of original deep CNN models for automated skin diseases classification.

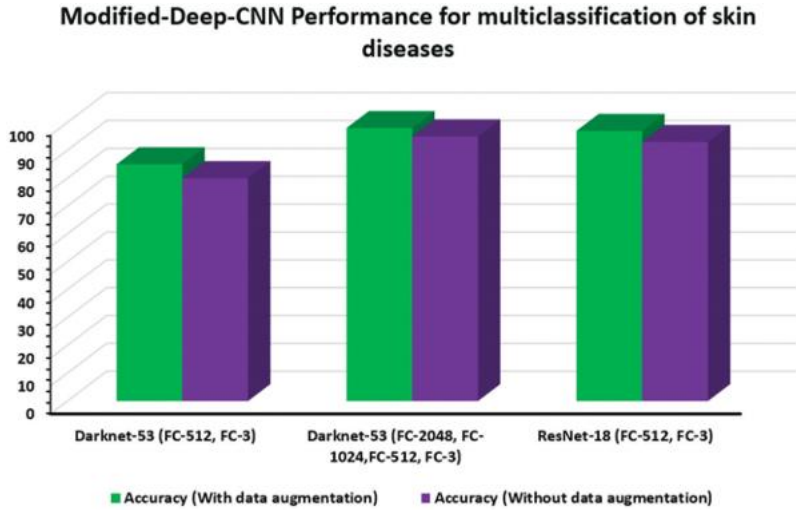


Figure 15. Performance of modified deep CNN models for automated skin diseases classification.

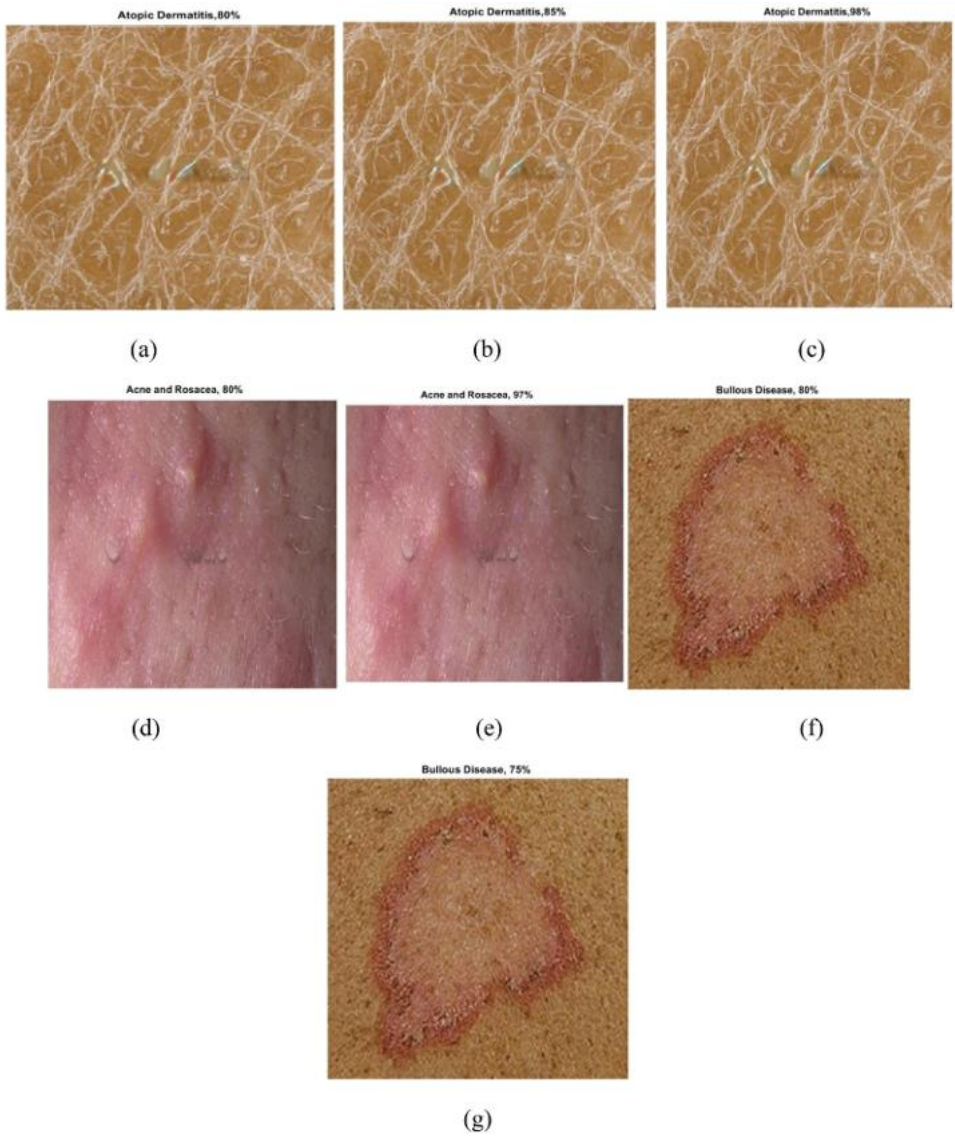


Figure 16. Representative examples of identification of skin disease by the different CNNs: (a) DarkNet-53, (b) DarkNet-53-2FC3, (c) DarkNet-53-4FC3, (d) ResNet-18, (e) ResNet-18-2FC3, (f) SqueezeNet, (g) EfficientNet-b0.

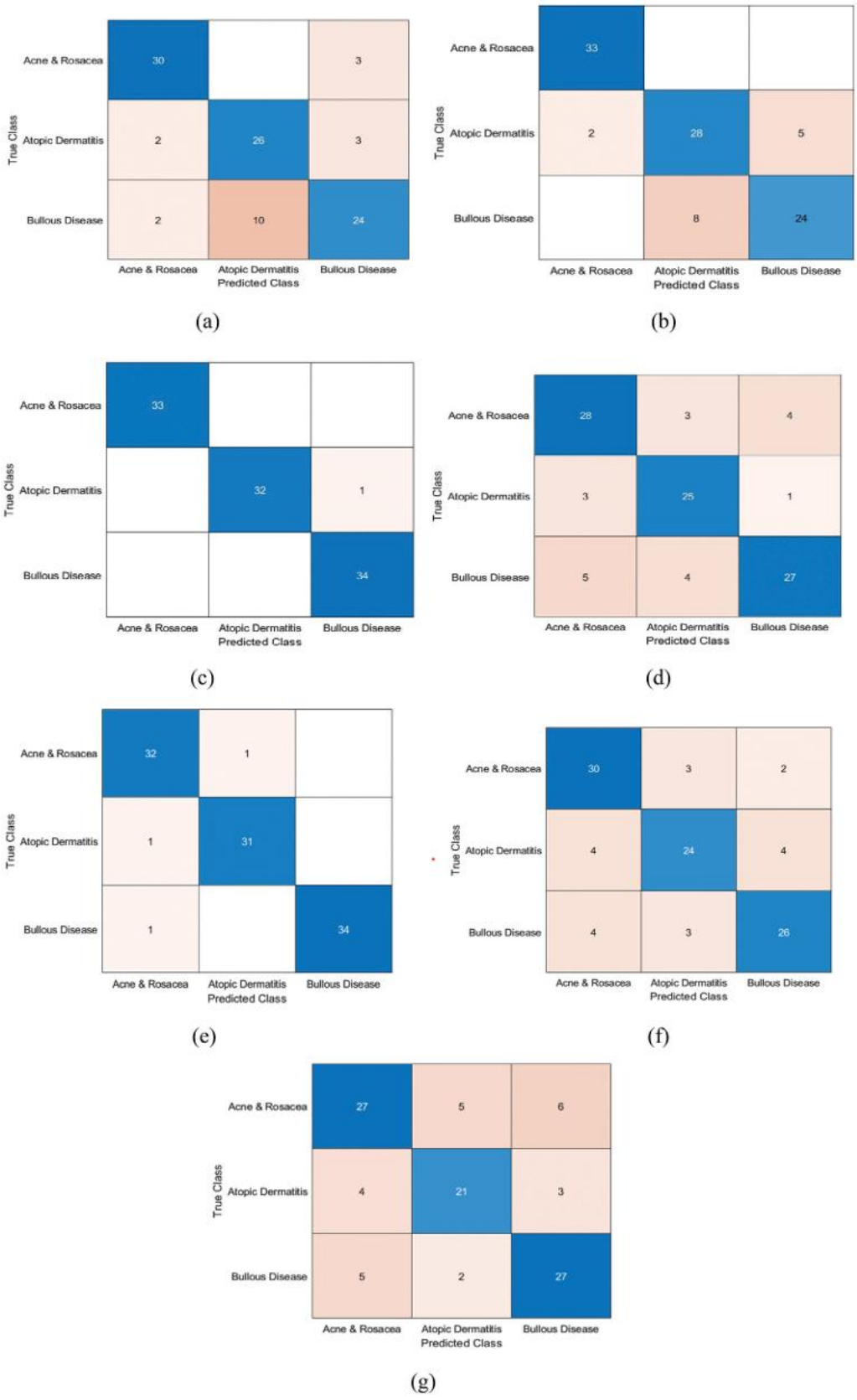


Figure 17. Confusion matrix obtained for the different CNNs: (a) DarkNet-53, (b) DarkNet-53-2FC3, (c) DarkNet-53-4FC3, (d) ResNet-18, (e) ResNet-18-2FC3, (f) SqueezeNet, (g) EfficientNet-b0.

Table 4. Performance metrics of original pre-trained CNNs.

Network	Accuracy (%)	Precision (%)	Recall (%)	Specificity (%)	F1-Score (%)	MCC (%)
DarkNet-53	80	80	80	90	80	82
ResNet-18	80	79	80	88	79	95
SqueezeNet	80	79	79	88	79	71
EfficientNet-b0	75	75	75	85	75	57

Table 5. Performance metrics of modified pre-trained CNNs.

Network	Accuracy (%)	Precision (%)	Recall (%)	Specificity (%)	F1-Score (%)	MCC (%)
DarkNet-53-2FC3	85	84	86	91	85	96
DarkNet-53-4FC3	98	99	98	99	98	99
ResNet-18-2FC3	97	96	96	98	96	93

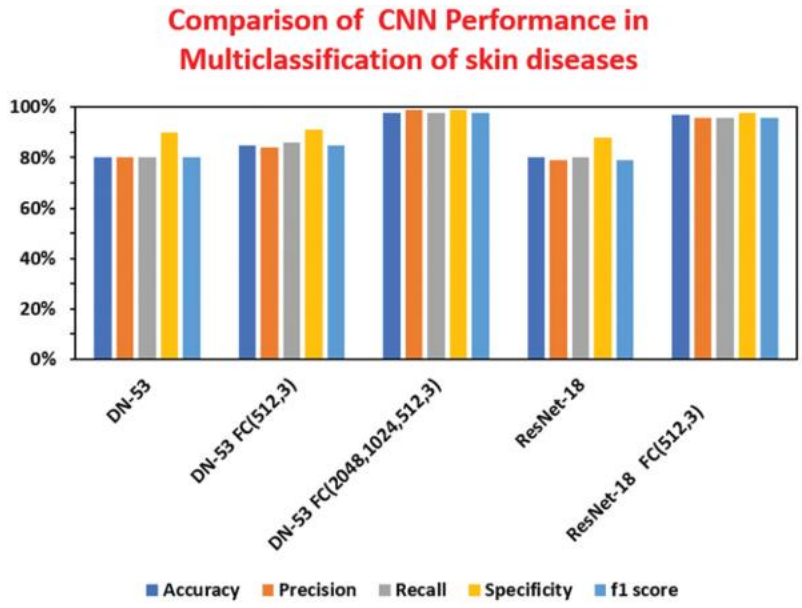


Figure 18. Performance metrics comparison of the original and modified CNNs.

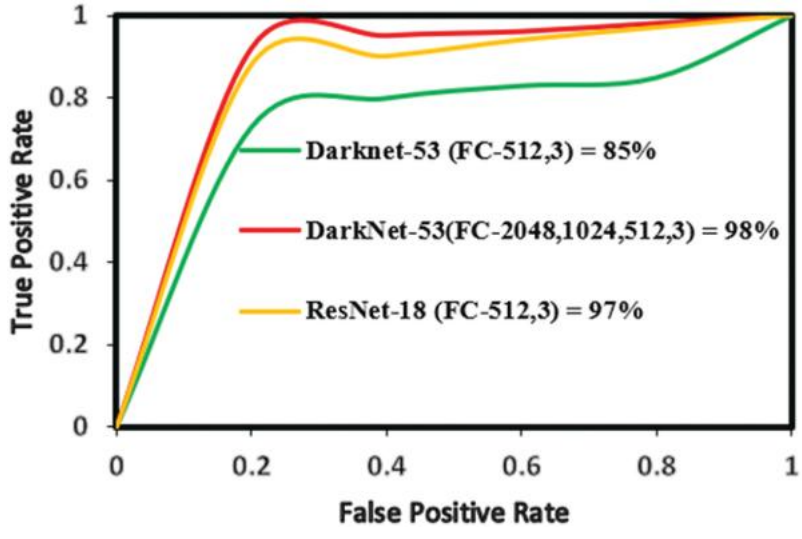


Figure 19. Comparison of ROC curves of the modified CNNs for automated identification of rosacea, atopic dermatitis and bullous disease.

comparative analysis in clinical settings will pave the way for practical implementation of these CNNs in aiding patient outcomes and development of treatment strategies.

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Author contributions

Both authors contributed to the conception and design. Dr G. Divya Deepak developed and trained the CNN models. Dr Subraya Krishna Bhat evaluated and analysed the performance metrics. The first draft of the manuscript was written by Dr G. Divya Deepak. It was revised by Dr Subraya Krishna Bhat and Dr G. Divya Deepak. Both authors read and approved the final manuscript.

Statement of informed consent

The dataset used in this study is from a public database. No identifying details are shown in the present paper.

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